WORKED SOLUTION

STEP III 2009, Question 6

Pure • Ptolemy's Theorem

> **STEP III 2009, Question 6**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/09-s3-q6

The key idea

The distance between two points on the unit circle depends only on the angle between them, and the trick for finding it is to pull out the "average" angle: $e^{i\beta} - e^{i\alpha}$ is $e^{i(\alpha+\beta)/2}$ times something purely imaginary. Once every chord is a sine of a half-angle difference, the big identity becomes a trigonometric identity, which product-to-sum formulae dispatch in a couple of lines. The four points then form a quadrilateral inscribed in a circle, and the identity is a famous theorem about it.

Solution

The chord length

Factor out $e^{i(\alpha+\beta)/2}$:

$$\begin{aligned} e^{i\beta} - e^{i\alpha} &= e^{i(\alpha+\beta)/2}\left(e^{i(\beta-\alpha)/2} - e^{-i(\beta-\alpha)/2}\right) \\ &= e^{i(\alpha+\beta)/2} \cdot 2i \sin \tfrac{1}{2}(\beta - \alpha). \end{aligned}$$

Since $|e^{i(\alpha+\beta)/2}| = 1$ and $|2i| = 2$,

$$|e^{i\beta} - e^{i\alpha}| = 2\,|\sin \tfrac{1}{2}(\beta - \alpha)|.$$

Finally, $0 < \alpha < \beta < 2\pi$ gives $0 < \frac{1}{2}(\beta - \alpha) < \pi$, so the sine is positive and the modulus signs can go: $|e^{i\beta} - e^{i\alpha}| = 2 \sin \frac{1}{2}(\beta - \alpha)$, as required.

> **Ask yourself** Why is the condition $\alpha < \beta$ there at all?

Without it, $\frac{1}{2}(\beta - \alpha)$ could lie in $(-\pi, 0)$ and the sine would be negative, so the formula would give a negative length. Ordering the angles is exactly what lets us drop the modulus. Note also that $|e^{i\alpha} - e^{i\beta}| = |e^{i\beta} - e^{i\alpha}|$, so the order inside the modulus doesn't matter.

The identity

Because $0 < \alpha < \beta < \gamma < \delta < 2\pi$, every pair appearing in the identity is in increasing order, so the first result applies to all six moduli. Dividing through by the common factor $2 \cdot 2 = 4$, we need to show

$$\sin\tfrac{1}{2}(\beta-\alpha)\sin\tfrac{1}{2}(\delta-\gamma) + \sin\tfrac{1}{2}(\gamma-\beta)\sin\tfrac{1}{2}(\delta-\alpha) = \sin\tfrac{1}{2}(\gamma-\alpha)\sin\tfrac{1}{2}(\delta-\beta).$$

The natural variables are the three gaps between consecutive points. Put

$$x = \tfrac{1}{2}(\beta-\alpha), \qquad y = \tfrac{1}{2}(\gamma-\beta), \qquad z = \tfrac{1}{2}(\delta-\gamma),$$

so that $\frac{1}{2}(\gamma-\alpha) = x+y$, $\frac{1}{2}(\delta-\beta) = y+z$ and $\frac{1}{2}(\delta-\alpha) = x+y+z$. The claim is

$$\sin x \sin z + \sin y \sin(x+y+z) = \sin(x+y)\sin(y+z).$$

Using $2\sin P \sin Q = \cos(P-Q) - \cos(P+Q)$ on each product:

$$\begin{aligned} 2\sin x \sin z &= \cos(x-z) - \cos(x+z), \\ 2\sin y \sin(x+y+z) &= \cos(x+z) - \cos(x+2y+z), \\ 2\sin(x+y)\sin(y+z) &= \cos(x-z) - \cos(x+2y+z). \end{aligned}$$

Adding the first two lines, the $\cos(x+z)$ terms cancel and we get exactly the third line. So the trigonometric identity holds (for all x, y, z, in fact), and multiplying back by 4 and using the chord formula gives

$$|e^{i\alpha} - e^{i\beta}|\,|e^{i\gamma} - e^{i\delta}| + |e^{i\beta} - e^{i\gamma}|\,|e^{i\alpha} - e^{i\delta}| = |e^{i\alpha} - e^{i\gamma}|\,|e^{i\beta} - e^{i\delta}|,$$

as required.

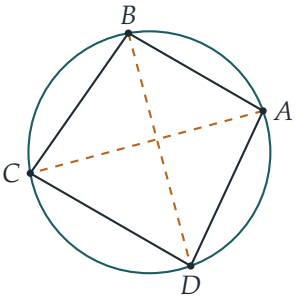
Ask yourself The trig identity holds for all x, y, z. So where did we use the ordering $\alpha < \beta < \gamma < \delta$?

In replacing each modulus by $2\sin(\text{half-difference})$ *without* modulus signs. If the points were out of order, some of those sines would be negative and the identity between the lengths would generally fail. For example, with angles $0.3, 2, 1, 4$ (so $\gamma < \beta$) the left side is about 4.84 and the right side about 1.15.

Interpretation

Let A, B, C, D be the points $e^{i\alpha}, e^{i\beta}, e^{i\gamma}, e^{i\delta}$. They lie on the unit circle, and the ordering of the angles means we meet them in the order A, B, C, D going anticlockwise. So $ABCD$ is a (convex) cyclic quadrilateral with sides AB, BC, CD, DA and diagonals AC, BD, and the identity says

$$AB \cdot CD + BC \cdot AD = AC \cdot BD.$$



Any circle is obtained from the unit circle by a scaling and a translation, and both sides of this equation scale by the same factor (the square of the scale factor), so the result holds for every cyclic quadrilateral: *in a cyclic quadrilateral, the sum of the products of the two pairs of opposite sides equals the product of the diagonals.* This is Ptolemy's theorem.

Checks

- A square inscribed in the unit circle ($\alpha, \beta, \gamma, \delta = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$) has sides $\sqrt{2}$ and diagonals 2: $\sqrt{2}\cdot\sqrt{2} + \sqrt{2}\cdot\sqrt{2} = 4 = 2\cdot 2$.
- A rectangle with sides a, b gives $a^2 + b^2 = (\text{diagonal})^2$: Ptolemy contains Pythagoras.
- Numerically, with 10^5 random ordered quadruples of angles the two sides agreed to within 2×10^{-15}.

Common slips

- Writing $|e^{i\beta} - e^{i\alpha}| = 2\sin\frac{1}{2}(\beta - \alpha)$ without checking the sine is positive.
- Expanding $\sin\frac{1}{2}(\delta - \alpha)$ and friends with compound-angle formulae in terms of the original four angles: it works, but is far messier than using the three gaps x, y, z.
- Forgetting that the identity needs the vertices in cyclic order: it is the diagonals, not any two chords, on the right-hand side.

Beyond STEP

For four points in general position the same quantities satisfy only Ptolemy's *inequality* $AB\cdot CD + BC\cdot AD \geq AC\cdot BD$, with equality exactly when the points are concyclic in that order; a slick proof uses the complex identity $(a-b)(c-d) + (a-d)(b-c) = (a-c)(b-d)$ and the triangle inequality. Behind this lies the cross-ratio, an invariant of Möbius transformations, which sends circles to circles and is central to complex geometry. Needham's *Visual Complex Analysis* develops these ideas beautifully.

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